

PERFORMANCE WORK STATEMENT (PWS)

Naval Medical Research Unit Europe, Africa, Central [NAMRU EURAFCENT] Clinical evaluation of the Xpert® tropical fever test on the Genexpert® Edge X system using venous and capillary whole blood specimens and Komfo Anokye Teaching Hospital [KATH]

1. GENERAL:

This is a non-personnel services contract to provide sample collection and testing for protocol 258C2: Clinical evaluation of the Xpert® tropical fever test on the Genexpert® Edge X system using venous and capillary whole blood specimens. The Government shall not exercise any supervision or control over the contract service providers performing the services herein. Such contract service providers shall be accountable solely to the Contractor who, in turn is responsible to the Government.

1.1 Description of Services/Introduction: The contractor shall provide all personnel, equipment, supplies, facilities, transportation, tools, materials, supervision, and other items and non-personal services necessary to perform sample collection and testing for protocol 258C2 as defined in this Performance Work Statement except for those items specified as government furnished property and services. The contractor shall meet the standards in this contract.

1.2 Background: PERFORMANCE WORK STATEMENT (PWS) BETWEEN NAVAL HEALTH RESEARCH CENTER [NHRC] & NAVY MEDICAL RESEARCH UNIT - EURAFCENT [NAMRU EURAFCENT] FOR 258C2: Clinical evaluation of the Xpert® tropical fever test on the Genexpert® Edge X system using venous and capillary whole blood specimens is approved to conduct research at University of Ghana Medical Centre (UMGC). The Xpert Tropical Fever test, performed on the GeneXpert Instrument Systems, is an automated, multiplex real-time PCR assay intended for the qualitative in vitro identification of nucleic acids from multiple fever-causing pathogens: Plasmodium species (including differentiation and reporting of P. falciparum), Dengue virus, Zika virus, Chikungunya virus, and Leptospira species. The Xpert Tropical Fever test uses ethylenediaminetetraacetic acid (EDTA) VWB and CWB specimens collected by finger stick capillary or venous blood draw from patients who are symptomatic for AFI without localizing signs. The test is intended as an aid in the diagnosis of parasitic, viral, or bacterial infections that cause AFI and results should be used in conjunction with clinical and laboratory findings, in addition to epidemiological risk factors such as patient travel history.

1.3 Objectives: Evaluate the clinical performance of the Xpert Tropical Fever test using venous whole blood (VWB) and capillary whole blood (CWB) when performed in an environment representative of a Clinical Laboratory Improvement Amendment (CLIA)-waiver (CW) by untrained users on the GeneXpert Edge X System.

1.4 Scope: Collect capillary and venous blood samples from approximately 300 patients presenting with clinical suspicion of one of the fever causing pathogens for Tropical Fever (Malaria, Dengue, Zika, Chikungunya, and/or Leptospira) and history or residence in/or travel to a geographic region within the last 3 months in which Tropical Fever transmission occurs:

- i. 250 uL CWB collected in EDTA coated micro-collection device.
- ii. 2.0 mL VWB collected in EDTA coated blood collection tube. 2.0 mL.
- iii. VWB collected in EDTA coated blood collection tube.

- Test capillary blood on-site on the Edge X and GXDX.
- Perform laboratory analysis on Edge X, GXDX on-site on a portion of the venous blood and perform Biofire Global Fever Panel comparator testing and Discrepant analysis with use of bi-directional sequencing at sites performing lab (NMIMR)

Period of Performance: The period of performance shall be for one (1) Base Year of 12 months and one (1) 6-months option period.

1.6 General Information

1.6.1 Quality Control: N/A

1.6.2 Quality Assurance: The government shall evaluate the contractor's performance under this contract in accordance with the Quality Assurance Surveillance Plan. This plan is primarily focused on what the Government must do to ensure that the contractor has performed in accordance with the performance standards. It defines how the performance standards will be applied, the frequency of surveillance, and the minimum acceptable defect rate(s).

1.6.3 Recognized Holidays: The contractor is not required to work on the U.S. Federal Government holidays listed below or on the U.S. Embassy designated Ghanaian holidays.

New Year's Day	Labor Day
Martin Luther King Jr.'s Birthday	Columbus Day
President's Day	Veteran's Day
Memorial Day	Thanksgiving Day
Independence Day	Christmas Day

1.6.4 Hours of Operation: The contractor is responsible for conducting business between the hours of 0800-1600, Monday through Friday except official Government holidays or when the facility is closed due to local or national emergencies, administrative closings, or similar Government directed facility closings. **Each assigned role is expected to contribute between five (5) hours to twenty (20) per week to the project, as agreed between NHRC and Cepheid in the accompanying Cepheid Tropical Fever Ghana Sites Budget Combined FY26.** For other than firm fixed price contracts, the contractor will not be reimbursed when the government facility is closed for the above reasons. The Contractor must at all times maintain an adequate workforce for the uninterrupted performance of all tasks defined within this PWS when the Government facility is not closed for the above reasons. When hiring personnel, the Contractor shall keep in mind that the stability and continuity of the workforce are essential.

1.6.5 Place of Performance: The work to be performed under this contract will be performed at the Komfo Anokye Teaching Hospital.

1.6.6 Security Requirements: Contractor personnel performing work under this contract and working at Komfo Anokye Teaching Hospital (KATH) possess a basic security clearance for entry and work at KATH at time of the proposal submission and must maintain the level of security required for the life of the contract.

1.6. 6.1 PHYSICAL Security: The contractor shall be responsible for safeguarding all government equipment, information and property provided for contractor use. At the close of each work period, government facilities, equipment, and materials shall be secured.

1.6. 6.2 Key Control: The Contractor shall be responsible for key control and shall follow local regulations when implementing their key control measures.

1.6. 7 Post Award Conference/Periodic Progress Meetings: The Contractor agrees to attend any post award conference convened by the contracting activity or contract administration office in accordance with Federal Acquisition Regulation Subpart 42.5. The contracting officer, Contracting Officers Representative (COR), and other Government personnel, as appropriate, may meet periodically with the contractor to review the contractor's performance. At these meetings the contracting officer will appraise the contractor of how the government views the contractor's performance and the contractor will appraise the Government of problems, if any, being experienced. Appropriate action shall be taken to resolve outstanding issues. These meetings shall be at no additional cost to the government.

1.6. 8 Contracting Officer Representative (COR): The (COR) will be identified by separate letter. The COR monitors all technical aspects of the contract and assists in contract administration. The COR is authorized to perform the following functions: assure that the Contractor performs the technical requirements of the contract; perform inspections necessary in connection with contract performance; maintain written and oral communications with the Contractor concerning technical aspects of the contract; issue written interpretations of technical requirements, including Government drawings, designs, specifications; monitor Contractor's performance and notifies both the Contracting Officer and Contractor of any deficiencies; coordinate availability of government furnished property, and provide site entry of Contractor personnel. A letter of designation issued to the COR, a copy of which is sent to the Contractor, states the responsibilities and limitations of the COR, especially with regard to changes in cost or price, estimates or changes in delivery dates. The COR is not authorized to change any of the terms and conditions of the resulting order.

1.6. 9 Data Rights: To the extent that any portion of the Research Plan is funded in whole or in part by the U.S. Government, the Government shall have unlimited rights, as defined under applicable U.S. federal regulations including FAR 52.227-14 and DFARS 252.227-7013/7014, to use, reproduce, modify, distribute, perform, and display final reports, data deliverables, and documentation expressly required to be delivered to the Government under this Agreement, for U.S. Governmental purposes. For avoidance of doubt, this clause does not apply to Background IP, including pre-existing software, analytical methods, databases, protocols, or institutional research tools, which shall remain the property of the originating Party. Research data and Research Results generated under this Agreement shall remain jointly owned by the Parties, subject to the U.S. Government's perpetual, royalty-free, irrevocable, non-exclusive, worldwide license for Governmental purposes. Each Party agrees to grant and hereby grants to the U.S. Government a royalty-free, perpetual, non-exclusive, worldwide license to use, for academic, research, or U.S. Governmental purposes, any Inventions made under this Agreement. Nothing in this clause shall be construed to limit academic publication rights, alter inventorship under U.S. patent law, or override the confidentiality or publication provisions of this Agreement except as expressly stated herein.

1.6.10 Organizational Conflict of Interest: Contractor and subcontractor personnel performing work under this contract may receive, have access to or participate in the development of proprietary or source selection information (e.g., cost or pricing information, budget information or analyses, specifications or work statements, etc.) or perform evaluation services which may create a current or subsequent Organizational Conflict of Interests (OCI) as defined in FAR Subpart 9.5. The Contractor shall notify the Contracting Officer immediately whenever it becomes aware that such access or participation may result in any actual or potential OCI and

shall promptly submit a plan to the Contracting Officer to avoid or mitigate any such OCI. The Contractor's mitigation plan will be determined to be acceptable solely at the discretion of the Contracting Officer and in the event the Contracting Officer unilaterally determines that any such OCI cannot be satisfactorily avoided or mitigated, the Contracting Officer may affect other remedies as he or she deems necessary, including prohibiting the Contractor from participation in subsequent contracted requirements which may be affected by the OCI.

1.6.11 PHASE IN /PHASE OUT PERIOD: Initial training and equipment acceptance will be expected to last up to two months. It is estimated that this study will last approximately 12 months to collect the necessary number of samples. An additional two months is expected to last in finalizing reports and disposition of material.

2. DEFINITIONS AND ACRONYMS:

2.1. DEFINITIONS:

2.1.1. **CONTRACTOR.** A supplier or vendor awarded a contract to provide specific supplies or service to the government. The term used in this contract refers to the prime.

2.1.2. **CONTRACTING OFFICER.** A person with authority to enter into, administer, and or terminate contracts, and make related determinations and findings on behalf of the government. Note: The only individual who can legally bind the government.

2.1.3. **CONTRACTING OFFICER'S REPRESENTATIVE (COR).** An employee of the U.S. Government appointed by the contracting officer to administer the contract. Such appointment shall be in writing and shall state the scope of authority and limitations. This individual has authority to provide technical direction to the Contractor as long as that direction is within the scope of the contract, does not constitute a change, and has no funding implications. This individual does NOT have authority to change the terms and conditions of the contract.

2.1.4. **DEFECTIVE SERVICE.** A service output that does not meet the standard of performance associated with the Performance Work Statement.

2.1.5. **DELIVERABLE.** Anything that can be physically delivered, but may include non-manufactured things such as meeting minutes or reports.

2.1.6. **KEY PERSONNEL.** Contractor personnel that are evaluated in a source selection process and that may be required to be used in the performance of a contract by the Key Personnel listed in the PWS. When key personnel are used as an evaluation factor in best value procurement, an offer can be rejected if it does not have a firm commitment from the persons that are listed in the proposal.

2.1.7. **PHYSICAL SECURITY.** Actions that prevent the loss or damage of Government property.

2.1.8. **QUALITY ASSURANCE.** The government procedures to verify that services being performed by the Contractor are performed according to acceptable standards.

2.1.9. QUALITY ASSURANCE Surveillance Plan (QASP). An organized written document specifying the surveillance methodology to be used for surveillance of contractor performance.

2.1.10. QUALITY CONTROL. All necessary measures taken by the Contractor to assure that the quality of an end product or service shall meet contract requirements.

2.1.11. SUBCONTRACTOR. One that enters into a contract with a prime contractor. There is no direct contractual relationship between the Government and the subcontractor.

2.1.12. WORK DAY. The number of hours per day the Contractor provides services in accordance with the contract.

2.1.12. WORK WEEK. Monday through Friday, unless specified otherwise.

2.2. ACRONYMS:

ACOR	Alternate Contracting Officer's Representative
AFARS	Army Federal Acquisition Regulation Supplement
AR	Army Regulation
CCE	Contracting Center of Excellence
CFR	Code of Federal Regulations
CONUS	Continental United States (excludes Alaska and Hawaii)
COR	Contracting Officer Representative
COTR	Contracting Officer's Technical Representative
COTS	Commercial-Off-the-Shelf
DA	Department of the Army
DD250	Department of Defense Form 250 (Receiving Report)
DD254	Department of Defense Contract Security Requirement List
DFARS	Defense Federal Acquisition Regulation Supplement
DMDC	Defense Manpower Data Center
DOD	Department of Defense
FAR	Federal Acquisition Regulation
GDDRP	Global Disease Detection and Response Program
HIPAA	Health Insurance Portability and Accountability Act of 1996
KATH	Komfo Anokye Teaching Hospital
KO	Contracting Officer
MOH	Ministry of Health
NAMRU EURAFCENT Command	US Naval Medical Research Unit Europe, Africa, Central
OCI	Organizational Conflict of Interest
OCONUS	Outside Continental United States (includes Alaska and Hawaii)
ODC	Other Direct Costs
PI	Principal Investigator
PIPO	Phase In/Phase Out
POC	Point of Contact
PRS	Performance Requirements Summary
PWS	Performance Work Statement
QA	Quality Assurance
QAP	Quality Assurance Program
QASP	Quality Assurance Surveillance Plan
QC	Quality Control

QCP
TE
WHO EMRO

Quality Control Program
Technical Exhibit
World Health Organization Eastern Mediterranean Office

3. GOVERNMENT FURNISHED ITEMS AND SERVICES:

3.1. Services: Shipping/ Handling, Participant compensation; Ethics/ Regulatory approval (Including amendment, renewals); Site initiation and study closeout; Data access; Stationery and printing; Team, training, meetings and coordination; Storage of samples; Community Entry Fees; Administration and logistics

3.2 Facilities: N/A

3.3 Utilities: N/A.

3.4 Equipment: Through the Government, Cepheid will be providing the GenXPert machine, pouches to run on the GenXPert, and assorted supplies for collecting patient samples, running on the instrument, and shipping to Noguchi for further testing.

3.5 Materials: Through the Government Cepheid will provide the necessary materials for the contractor staff to provide the support outlined in the Protocol 258C2 and PWS for the work to be performed in the KATH clinical area. Stationery and printing, Personal Protective Equipment (PPE), will also be funded/ supplied. See attachment 7 258C2_Site 605_Packing List_Cepheid Komfo Anokye Teaching Hospital Equipment furnished by government

4. CONTRACTOR FURNISHED ITEMS AND RESPONSIBILITIES:

4.1 General: Lab space

4.2 Secret Facility Clearance: N/A

4.3. Materials:

Collected capillary and venous blood samples from approximately 100 patients presenting with clinical suspicion of one of the fever causing pathogens for Tropical Fever (Malaria, Dengue, Zika, Chikungunya, and/or Leptospira) and history or residence in/or travel to a geographic region within the last 3 months in which Tropical Fever transmission occurs.

- i. 250 uL CWB collected in EDTA coated micro-collection device.
- ii. 2.0 mL VWB collected in EDTA coated blood collection tube. 2.0 mL.
- iii. VWB collected in EDTA coated blood collection tube.

4.4. Equipment: N/A

5. Specific Tasks:

5.1. Basic Services. The contractor shall provide services for laboratory research as defined in the Tropical Fever Study Protocol.

- 5.1.1 Conduct study procedures as described on protocol 258C2 (refer to attachments).
- 5.1.2 Obtain Institutional Review Board (IRB) approval for protocol & Informed Consent Form (ICF), prior to study initiation and submit protocol amendments/deviations, progress reports, special reports to IRB.
- 5.1.3 Performing site IRBs will review and approve documentation, according to country specific regulations prior to trial initiation. Documentation reviewed may include but is not limited to: protocol, principal investigators, informed consents, recruitment procedures, all printed information given to subjects, and information and/or statements regarding compensation.
- 5.1.4 Performing site will be responsible for the proper conduct of the trial and should be familiar with the appropriate use of Good Clinical Practice (GCP) and applicable regulations.
- 5.1.5 Ensure adequate delegation of responsibilities and Cepheid led training occurs for all persons assisting with the trial on the protocol, the investigational product and their trial-related duties and functions.
- 5.1.6 Ensure all privacy and confidentiality safeguards are in place prior to study initiation, and throughout the study, as approved by the IRB.
- 5.1.7 Create and maintain a regulatory binder. The regulatory binder will stay at the performer site for the duration of the study; however, the study coordinator will update documents (i.e. protocol modifications, IRB-stamped documents, training certificates, written correspondence) as applicable.
- 5.1.8 Consent and enroll patients according to the IRB-approved protocol and following all privacy and confidentiality safeguards.
- 5.1.9 Prepare samples for device testing according to package insert instructions. Operate the GeneXpert and Edge X following manufacturer instructions.
- 5.1.10 Maintain laboratory and laboratory study equipment in conjunction with Safe Laboratory Practice (SLP) guidelines.
- 5.1.11 Perform Biofire Global Fever Panel comparator testing and discrepant analysis with use of bi-directional Sanger sequencing at the performing lab.
- 5.1.12 Communicate with NAMRU EURAFCENT regarding study specifics of their concern.
- 5.1.13 Provide reports to NAMRU EURAFCENT per deliverables to maintain project records (as appropriate), decisions and periodic status of project progress through:
- i. Periodic/regular teleconferences with NAMRU EURAFCENT and UMGC.
 - ii. Monthly site status reports.
 - iii. Site visit reports to be submitted subsequent to each site visit.
 - iv. Copies of all IRB protocols, amendments and associated documents and their English translations (if applicable).
 - v. Collection and testing of samples and data entry and reports provided to study monitor as instructed per protocol.
- 5.1.14 The following roles and activities are expected to support the facilitation of the protocol

- i. Principal Investigator (PI) oversight during the twelve month study window
- ii. Co-PI 1 oversight during the twelve month study window
- iii. Co-PI 2 oversight during the twelve month study window
- iv. Site Administrator will spend an average of five hours per week for the study duration
- v. Phlebotomist for sampling (phlebotomy and finger pricking) of both adult and pediatric patients. Four personnel allows for continuous coverage.
- vi. Consenter to be split between adult (2) and pediatric (2) participants
- vii. Medical Screener to be split between adult (2) and pediatric (2) participants
- viii. BioFire Operators, Two to ensure continuity
- ix. Data entry personnel, Two data entry personnel at 100% FTE each, one for primary data entry, another for quality assurance
- x. Research Assistants to assist with sample processing, storage and retrieval

5.2. Task Heading: N/A

5.3. CONTRACTOR MANAGEMENT REPORTING (CMR): N/A

6. APPLICABLE PUBLICATIONS (CURRENT EDITIONS; N/A)

6.1. The Contractor must abide by all applicable regulations, publications, manuals, and local policies and procedures.

6.2 The Contractor must abide by all procedures defined in Protocol 258C2: Clinical Evaluation of the Xpert® Tropical Fever Test on the Genexpert® Edge X System Using Venous and Capillary Whole Blood Specimens.

7. Attachment/Technical Exhibit List:

7.1. Attachment 1/Technical Exhibit 1 – Performance Requirements Summary *(This document is required for every PWS. See attached example for format.)*

7.2. Attachment 2/Technical Exhibit 2 – Deliverables Schedule *(This document is required for every PWS. See attached example for format.)*

7.3 Attachment 3/Technical Exhibit 3 – Estimated Workload Data *(This document should be made available if historical data exists. See attached example for format.)*

7.4 Attachment 4 - Collaborative Data Use Statement

7.5 Protocol 258C2: Clinical Evaluation of the Xpert® Tropical Fever Test on the Genexpert® Edge X System Using Venous and Capillary Whole Blood Specimens.

7.6 Statement of Work Between Naval Health Research Center [NHRC] and Naval Medical Research Unit – EURAFCENT [NAMRU EURAFCENT] for 258C2: Clinical evaluation of the Xpert® tropical fever test on the Genexpert® Edge X system using venous and capillary whole blood specimens

7.7 CORRECTED Cepheid Tropical Fever Ghana Sites Budget Combined FY26

7.8 258C2_Site 605_Packing List_Cepheid Komfo Anokye Teaching Hospital Equipment
furnished by government

7.8 Material Transfer Agreement (MTA) _Xpert Tropical1

TECHNICAL EXHIBIT 1

Performance Requirements Summary

The contractor service requirements are summarized into performance objectives that relate directly to mission essential items. The performance threshold briefly describes the minimum acceptable levels of service required for each requirement. These thresholds are critical to mission success.

Performance Objective (The Service required—usually a shall statement)	Standard	Performance Threshold (This is the maximum error rate. It could possibly be “Zero deviation from standard”)	Method of Surveillance
PRS # 1 <i>The contractor shall</i> Conduct study procedures as described on protocol 258C2	Conduct study in accordance with signed Statement of Investigator, study protocol, Good Clinical Practices, Good Laboratory Practices, Safe Laboratory Practice (SLP) guidelines, and other applicable FDA and/or country specific regulations, regulatory requirements, the Belmont Report, and the Declaration of Helsinki	Compliance shall be maintained at a rate of not less than 98%, as verified through periodic reviews or audits.	Periodic Surveillance as defined by protocol 258C2
PRS # 2 <i>The contractor shall</i> attend Cepheid led training for all people assisting with the trial on the protocol, the investigational product and their trial-related duties and functions.	The contractor shall ensure that all technical staff maintain current training and competency certifications required from Cepheid as defined in protocol 258C2	100% training compliance for all Cepheid investigators and technicians	Periodic Surveillance
PRS # 3 <i>The contractor shall</i> Ensure all privacy and confidentiality safeguards are in place prior to study initiation, and throughout the study.	Approved by the IRB.	Zero deviation from the approved IRB standard	<i>100% at the beginning of the study</i>
PRS # 4 <i>The contractor shall</i> Create and maintain a regulatory binder at each study site.	Defined documentation found in protocol 258C2. The study coordinator will update documents (i.e. protocol modifications, IRB-stamped documents, training certificates, written correspondence) as applicable.	Compliance shall be maintained at a rate of not less than 98%, as verified through periodic reviews or audits	<i>Periodic Surveillance</i>
PRS # 5 <i>The contractor shall</i> obtain Consent and enroll patients	according to the IRB-approved protocol and follow all privacy and confidentiality safeguards.	<i>100% Compliance</i>	<i>Periodic Surveillance</i>

PRS # 6 <i>The contractor shall</i> Collect and Prepare samples for device testing	Operate the GeneXpert and Edge X following manufacturer instructions.	<i>95% success rate (valid results with no instrument/ operational errors) with 100% trained operator compliance</i>	<i>Periodic Surveillance</i>
PRS # 7 <i>The contractor shall</i> Maintain laboratory study equipment	In accordance with Safe Laboratory Practice (SLP) guidelines. Notify NAMRU EURAFCENT in equipment malfunctions, errors, damage.	<i>>98% operational uptime and 100% adherence to safe laboratory use</i>	<i>Periodic Surveillance</i>

TECHNICAL EXHIBIT 2

DELIVERABLES SCHEDULE

(This technical exhibit lists any reports or documentation that is required as a deliverable to include the frequency, # of copies, medium/format and who/where it is to be submitted. A deliverable is anything that can be physically delivered but may include non-physical things such as meeting minutes. Note: All PWS deliverables should be included in this exhibit.)

<u>Deliverable</u>	<u>Frequency</u>	<u># of Copies</u>	<u>Medium/Format</u>	<u>Submit To</u>
Execute Material Transfer Agreements	Prior to any shipment of biological samples to Noguchi Memorial Institute of Medical Research	1 printed copy to be filed in regulatory binder 1 electronic copy.	Accepted Data Transfer Agreement 1 printed copy to be filed in regulatory binder. 1 electronic copy/	IRB, NAMRU EURAFCENT Principal Investigator
Define Data Use Terms	Prior to executing the study	1 printed copy to be filed in regulatory binder 1 electronic copy	See Technical Exhibit 4	IRB, NAMRU EURAFCENT Principal Investigator
Obtain Institutional Review Board (IRB) approval for protocol & Informed Consent Form (ICF), and submit protocol amendments/deviations, progress reports, special reports to IRB	prior to study initiation	1 printed copy to be filed in regulatory binder 1 electronic copy.	1 printed copy to be filed in regulatory binder. 1 electronic copy.	IRB, NAMRU EURAFCENT Principal Investigator Required for Cepheid Review
Collect capillary and venous blood samples from approximately 100 patients presenting with clinical suspicion of one of the fever causing pathogens for Tropical Fever (Malaria, Dengue, Zika, Chikungunya,	For the duration of the 12-month study collection period	appx 100 samples	i.250 uL CWB collected in EDTA coated micro-collection device. ii. 2.0 mL VWB collected in EDTA coated blood collection tube.2.0 mL. iii. VWB collected in EDTA coated blood collection tube.	Noguchi Memorial Institute of Medical Research (NMIMR)

<u>Deliverable</u>	<u>Frequency</u>	<u># of Copies</u>	<u>Medium/Format</u>	<u>Submit To</u>
and/or Leptospira) and history or residence in/or travel to a geographic region within the last 3 months in which Tropical Fever transmission occurs.				
Test samples collected on site using GenXPert	For the duration of the 12-month study collection period	<i>Appx. 100 samples</i>	Aliquot	Noguchi for further advanced testing (sequencing).

TECHNICAL EXHIBIT 3
ESTIMATED WORKLOAD DATA

ITEM	NAME	ESTIMATED QUANTITY	
1	Principal Investigator/Consultant Emergency Physician	1	17 Months
2	Co-Investigator/Microbiologist	1	13 Months
3	Co-Investigator/Emergency Physician (CW User # 1)	1	14 Months
4	Co-Investigator/Pediatrician	1	13 Months
5	Emergency Medicine Physician # 2 (CW User #2)	1	13 Months
6	Screener/Consenter (Pediatrician)	1	13 Months
7	Screener/Consenter (Pediatrician)	1	13 Months
8	Triage Nurse (Screener, Emergency Medicine)	1	13 Months
9	Triage Nurse (Screener, Emergency Medicine)	1	13 Months
10	Triage Nurse (Screener, Pediatrics)	1	13 Months
11	Triage Nurse (Screener, Pediatrics)	1	13 Months
12	Phlebotomist (Emergency Medicine)	1	13 Months
13	Phlebotomist (Pediatrics)	1	13 Months
14	Lab Microbiology	1	14 Months
15	Lab Microbiology	1	13 Months
16	Research assistant (Data Entry)	1	15 Months
17	Research assistant (QA)	1	13 Months
18	Study Coordinator	1	17 Months
19	Finance Manager	1	14 Months

INVOICES AND PAYMENT

TECHNICAL POC:

CDR Christina Jamros

Email: christina.m.jamros.mil@health.mil

Alternate TPOC:

Dr. Naiki Attram

E-mail: attramn@state.gov

Dr. Shirley Nimo Paintsil

E-mail: nimopaintsilsc@state.gov

Courtesy copy to: Sherine Wahba Commercial#: +2-0122-7377142

E-mail: sherine.k.wahba.ln@health.mil

ADDRESS: NAMRU-EURAFCENT

USAID building - Takseem Laselki Maadi, Cairo, Egypt

The contractor shall submit a manually created invoice to the TPOC cited above. The TPOC will then review, prepare and submit the invoice package and send it to the payment office identified in block 18a.

Vendor invoice shall include the following information for proper invoice:

- Invoice Number
- Invoice Date
- Contract Line Item Number
- Service/Material Receipt Date
- Service/Material Acceptance Date

ATTACHMENT 4

COLLABORATIVE DATA USE TERMS

The terms of Collaborative Data Use is applied to support the research described in the study “Clinical Evaluation of the Xpert® Tropical Fever Test on the GeneXpert® Edge X System Using Venous and Capillary Whole Blood Specimens”

1. Definitions:

1.1. Research Resources: Research Resources is defined as the following data collected under Cepheid Protocol 258C2: Clinical evaluation of the Xpert® Tropical Fever Test on the Genexpert® Edge X system using venous and capillary whole blood specimens. (Study).

Study data: Consists solely of de-identified data collected on the electronic case report form (eCRFs). No direct identifiers or protected health information (PHI) are included. No data are extracted from patient medical records, electronic health records, or any other clinical documentation. Study information to be collected:

- Sex at birth
- Age range
- Race and ethnicity
- Clinically assessed body temperature and noted anatomical location of measured temperature at time of specimen collection
- Symptom(s) occurring within 10-days prior to specimen collection (self-reported and/or study staff assessed)
- Number of days since onset of symptom(s) (self-reported)
- Pregnancy status (self-reported)
- Antibiotics and/or antimalarials use (oral or intravenous) within three days prior to specimen collection (self-reported)
- Flavivirus vaccination history within 30-days prior to specimen collection (self-reported)
- Date and time of specimen collection

1.2. Limited Data Set (LDS): LDS means a limited set of identifiable data obtained from participants enrolled in Study which may be disclosed for research, public health and health care operations as set forth in 45 CFR 164.514(e)(2) and in Appendix B. The Limited Data Set includes the complete Limited Data Set and any portion, part or subset of the Limited Data Set and any and all copies thereof.

KATH (hereinafter referred to as “Provider”) will transfer Research Resources identified above to NAMRU EURAFCENT. No additional individual identifiers will be transferred to NAMRU EURAFCENT.

2. Terms of Data Use:

2.1. Use/Non-use of Research Resources. Recipient and Recipient Investigator agree that Research Resources will be solely used for Recipient’s conduct of the Research Plan. If Recipient or Recipient Investigator wishes to carry out research beyond that described in the Research Plan, such research will be carried out only pursuant to a further written contract, or an amendment to this Contract, signed by both Parties. Recipient further agrees not to transfer Provider’s Research Resources to any other party.

Recipient and Recipient Investigator further agree that the:

- i. Research Resources are to be used solely for Recipient's internal non-commercial research purposes of validating the Tropical fever test as stated in the protocol and will not be used in any research that is subject to consulting or licensing obligations to any for-profit organizations;
- ii. Research Resources will not be used in human subjects research, in clinical trials, or for diagnostic purposes involving human subjects without the written consent of the Provider(s);
- iii. Research Resources will be used only at the Recipient organization and only in the Recipient Investigator's laboratory under the direction of the Recipient Investigator or others working under his/her direct supervision; and
- iv. Research Resources will not be transferred to anyone else within the Recipient organization or study sponsor without the prior written consent of the Provider(s) or as specified in this Contract and in the executed SOW between NHRC and NAMRU EURAFCENT.
- v. Unauthorized uses or disclosures of LDS may constitute a 'breach' for breach notification rule purposes under the 1996 U.S. Health Insurance Portability and Accountability Act ("HIPAA");
- vi. Neither Recipient nor Recipient's Investigator will use LDS either alone or in conjunction with any other information in any effort whatsoever to establish the individual identities of any of the study participants from whom LDS was obtained.

2.2. Intellectual Property (IP)

2.2.1. Background IP. Provider(s) retain all intellectual property rights to its Research Resources, information and data as well as or potential future rights, such as issued patents, patent applications or invention disclosures ("Background IP"). No Party obtains any rights to the other Party's Background IP under this Contract except for the purpose of conducting the Research Plan.

2.2.2. Inventions. Ownership of any invention patentable under U.S. patent law which is conceived or first actually reduced to practice under this Contract ("Invention") will follow inventorship in accordance with U.S. patent law. Any Invention made jointly by the Parties under this Contract ("Joint Invention") will be owned jointly by the Parties. The Parties shall agree on filing, prosecution, maintenance, enforcement and defense for patent applications on each Joint Invention.

2.2.3. Patent Applications. The Parties shall promptly notify each other in writing of their intent on filing a patent application on Joint Invention(s) and agree to discuss the obligations and responsibilities of each Party with respect to the filing and prosecution of the patent application prior to the filing of any such patent application.

2.3. Confidential Information. Recipient agrees to maintain all proprietary information disclosed by a Provider(s) to the Recipient ("Confidential Information" or "CI") under this Contract, except to the extent such information:

- i. can be demonstrated to have been in the public domain or publicly known and readily available to the trade or the public prior to the date of the disclosure to Recipient;

- ii. can be demonstrated, from written records, to have been in Recipient's possession or readily available to Recipient from another source not under obligation of secrecy to Provider(s) prior to the disclosure;
- iii. becomes part of the public domain or publicly known by publication or otherwise, not due to any unauthorized act by Recipient; or
- iv. is independently developed by Recipient without reference to information provided by Provider(s).
- v. is required to be disclosed by law, regulation or court order.

2.3.1. If Recipient is required by law to disclose the CI including without limitation by discovery, subpoena or other legal or administrative process, Recipient agrees to provide Provider(s) prompt advance notice of the required disclosure to permit Provider(s) at its option and expense, to seek an appropriate protective order or waive the requirements under this Contract. If no protective order or waiver is obtained and disclosure is legally required, such disclosure may be made but only to the extent required by law. Notwithstanding the foregoing, in recognition of the fact that representatives of **NAMRU-EURAFCENT**, the U.S. Government, and the U.S. military are engaged in an active collaboration as part of the **Cepheid Protocol 258C2: Clinical evaluation of the Xpert® Tropical Fever Test on the Genexpert® Edge X system using venous and capillary whole blood specimens**, **NAMRU-EURAFCENT** as a Recipient of CI may further distribute CI to colleagues within **study sponsor** so long as these colleagues are made aware of the existence and terms of this Contract.

2.4. Publication of Research Results. All research data and other recorded information first produced in the conduct of the Research Plan ("Research Results") will be jointly owned by the Parties. The Parties will promptly inform each other by written communication, at mutually agreed upon timeframes, of Research Results. The Parties shall afford each other the opportunity to review and comment on any intended public disclosure associated with Research Results at least thirty (30) days in advance of any written submission (abstract or manuscript) or oral presentation that makes reference to the Research Results. Each Party shall delete Confidential Information of the other Party, if requested by that Party, from such submission or presentation and if requested by the other Party will withhold such submission or presentation for an additional thirty (30) days to allow an assessment of patentability. Upon the written request of a Party, the other Parties agree to delay submission or presentation for an additional thirty (30) days to allow for the filing of any relevant patent applications. It is intended that any public disclosure of Research Results be jointly authored by the Parties (as appropriate) in accordance with their respective scientific contributions. Recipient will acknowledge Provider(s) and any third-party contributors, as appropriate, as the source of the Research Resources in all publications containing any data or information derived from the Research Resources unless Provider(s) indicate otherwise in writing. Each Party agrees to hold the other Party's submission and Research Results in confidence until published and/or protected with a patent application as described in Paragraph 3.

2.4.1. LDS Publications. For publications related to LDS, study participant identifiers for LDS will not be used in any publications or presentations. Any such publication will include a) an acknowledgment that the project from which the Limited Data Set was derived was funded in whole or in part with Federal funds from Cooperative Contract issued by **JPEO through NHRC and NAMRU EURAFCENT**; and b) a statement that the content of the information does not necessarily reflect the position or the policy of the Government, and no official endorsement should be inferred.

2.5. No Warranties. Research Resources are provided “as is” without any representations or warranty of any kind, either expressed or implied, as to any matter whatsoever including, without limitation, warranties of merchantability and fitness for a particular use. Without limitation of the foregoing generality, nothing contained herein shall be construed as extending any representation or warranty, express or implied, with respect to the research conducted using the LDS or Research Results to be obtained hereof or that use of the LDS and/or the Research Results will not infringe any patent or other proprietary right.

2.6. Liability. Each Party is responsible for any damages it incurs as a result of its activities under this study. Recipient assumes all liability for damages that may arise in any way relating to Recipient’s possession and use of LDS, except to the extent permitted by law for any such claims or damages (including attorney’s fees) arising out of gross negligence, recklessness, or willful misconduct of Provider(s).

2.7. Indemnification. To the extent permitted by applicable law, **NAMRU EURAFCENT** will indemnify and hold Provider(s) and the U.S. Government harmless from and be responsible for any liability for, and will defend from any claims, costs, or damages or expenses, including attorney’s fees, resulting from, arising out of or in any way relating to **NAMRU EURAFCENT** possession and use of Research Resources, except for any such claims or damages (including attorney’s fees) arising out of gross negligence, recklessness, or willful misconduct of Provider(s). The liability of **NAMRU EURAFCENT** for any breach of this Contract, for any claims, costs, damages or expenses, including attorney’s fees, resulting from, arising out of **NAMRU EURAFCENT** possession and use of Research Resources, for any negligence, or arising in any other way out of the subject-matter of this Contract will not extend to any indirect, consequential, or punitive damages or losses, or to any loss of profits, loss of revenue, loss of business, loss of data, loss of contracts or opportunity, whether direct or indirect.

2.8. Force Majeure. No Party will be liable for any unforeseeable event beyond its reasonable control and not caused by its own fault or negligence, which causes the Party to be unable to perform its obligations under this contract, and which it has been unable to overcome by the exercise of due diligence. If a force majeure event occurs, the Party unable to perform will promptly notify the other Party and will use its best efforts to resume performance as quickly as possible and will suspend performance only for such period of time as is necessary as a result of the force majeure event.

2.9. Compliance with Laws, Regulations and Institutional Policies. Recipient and Recipient’s Investigator shall comply with all applicable laws, rules and regulations including, without limitation, all applicable current governmental regulatory requirements, concerning the use of LDS. Recipient further agrees to establish and maintain appropriate administrative, technical, and physical safeguards to protect the confidentiality of the LDS and to prevent unauthorized use or access to it. In addition, Recipient and Recipient’s Investigator represent that the conditions for use of Research Resources have been approved by Recipient’s IRB or equivalent body, in accordance with applicable law including but not limited to 45 CFR Part 46 and 21 CFR Parts 50 and 56.

2.10. Term and Termination. The contract shall begin on the Effective Date and shall expire on the earliest of the following dates: (a) upon completion of the Research Plan, or (b) on thirty (30) days written notice by a Party to the other Party, or (c) failure to agree on mutually agreeable terms for cure of a breach of material term within this Contract within thirty (30) days of notice of breach. Upon termination of the Contract, Recipient will discontinue its use of Research

Resources and will, upon direction of Provider(s), return or destroy any remaining Research Resources and all copies thereof.

2.11. Governing Law. The use of data shall be governed by Ghanaian and U.S. law, without regard to its principles of conflicts of law.

2.12. Surviving Provisions. Paragraphs 2.2 through 2.9, 2.13 and 2.14 survive the expiration or early termination of this Contract.

2.13. Amendments. Amendments to the definition of Data Use must be made in writing and signed by authorized representatives of both Parties.

2.14. Assignment: The contract shall be binding on the Parties hereto and upon their respective heirs, administrators, successors and permitted assigns. This contract may not be assigned by either Party or by operation of law without the prior written consent of and authorized individual of the other Party.

2.15. Independent Parties: The Parties to this contract are independent contractors and not agents of the other. This contract shall not constitute a partnership or joint venture, and no Party may be bound by the other to any contract, arrangement or understanding except as specifically stated herein.

2.16. Counterparts: This contract may be executed simultaneously in one or more counterparts, each of which shall be deemed to be original but all of which together shall constitute one and the same Contract.

Notices: Formal notices required or permitted hereunder shall be given in writing. Written notices may be delivered personally, sent by a nationally/internationally recognized courier service or by first class mail, or transmitted electronically by, for example, or e-mail. All notices or communications required or permitted hereunder shall be deemed to have been given (a) if by personal delivery to the proper address and with receipt acknowledged, on the date of such delivery; (b) if by overnight courier service to the proper address and with receipt acknowledged, on the second business day following deposit if delivered; (c) if sent by e-mail, with confirmed transmission, on the next business day following transmission; or (d) if mailed, postage prepaid first-class certified or registered mail, return receipt requested, on the fifth business day after mailing, to the address designated below or to such other address as a party may designate to the other parties in writing. Any Party may change their address for the purposes of this Contract by giving written notice in accordance with this Paragraph. Any notices related to this Contract shall be sent as follows:

If to **NAMRU-EURAFCENT**:

Naval Air Station II Sigonella B476 Strada Statale 417 Catania Sigonella, Italy 95030
Attn: Cdr Christina Jamros, Deputy Science Director
Email: christina.m.jamros.mil@health.mil

With a Copy to:

Attn: Sherry Selim, Support Contract Manager
Phone: +20 12 234 69588

Email: sherry.n.selim.ln@health.mil

If to **KATH**:

Attn: Medical Director
Komfo Anokye Teaching Hospital,
Okomfo Anokye Road, Kumasi, Ghana

Copies of manuscripts and/or abstracts sent for review and comment by

NAMRU-EURAFCENT pursuant to this Paragraph shall be sent to:

NAMRU-EURAFCENT:

Cdr. Christina Jamros, MC, USN
Email: christina.m.jamros.mil@health.mil

And

Tiziana Lemma, Medical Editor
tiziana.lemma.ctr@health.mil

With a copy to:

Copies of manuscripts and/or abstracts sent for review and comment by KATH and pursuant to this Paragraph shall be sent to:

KATH: George Oduro, MD
Email: gdoduro@hotmail.com